

Q6: Explain how image formation models help in improving image quality in applications such as autonomous driving.

Introduction

Image formation is the process by which a real-world scene is captured by a camera and converted into a digital image. An image formation model describes how light from objects passes through the camera lens, reaches the image sensor, and forms the final image. In autonomous driving, image formation models are essential because vehicles rely on high-quality images to detect roads, pedestrians, traffic signs, and obstacles accurately.

How Image Formation Models Improve Image Quality

1. Proper Light Capture

- The camera lens collects light reflected from objects.
- Image formation models ensure that sufficient light reaches the image sensor.
- This produces brighter and clearer images, even under varying lighting conditions.

2. Lens Distortion Correction

- Camera lenses may introduce distortions such as barrel or pincushion distortion.
- Image formation models help identify and correct these distortions.
- Straight roads and lane markings appear accurate after correction.

3. Noise Reduction

- Images captured in low-light or rainy conditions often contain sensor noise.
- Image formation models help reduce this noise using image enhancement techniques.
- This improves image clarity and object visibility.

4. Brightness and Contrast Enhancement

- Images captured in fog, shadows, or bright sunlight may have poor contrast.

- Image formation models adjust brightness and contrast to make objects easier to detect.

5. Camera Calibration

- Camera calibration determines the camera's internal and external parameters.
- It removes perspective errors and provides accurate measurements of object positions and distances.

6. Depth Estimation

- Stereo cameras use image formation models to estimate the distance between the vehicle and surrounding objects.
- This helps the vehicle maintain a safe distance and avoid collisions.

7. Accurate Object Detection

- Improved image quality allows the system to detect:
 - Lane markings
 - Traffic signs
 - Pedestrians
 - Other vehicles
 - Road obstacles
- Accurate detection improves driving safety and decision-making.

Applications in Autonomous Driving

- Lane detection
- Traffic sign recognition
- Pedestrian detection
- Vehicle detection
- Collision avoidance
- Parking assistance

- Road obstacle detection

Conclusion

Image formation models play a vital role in autonomous driving by producing clear and accurate digital images. They improve image quality through lens distortion correction, noise reduction, brightness and contrast enhancement, camera calibration, and depth estimation. These improvements enable reliable object detection, accurate navigation, and safer autonomous driving.